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Operations Research: Sensitivity Analysis

SENSITIVITY ANALYSIS

Reduced Cost

For any nonbasic variable, the reduced cost for the variable is the amount by which the nonbasic variable's objective function coefficient must be improved before that variable will become a basic variable in some optimal solution to the LP.

If the objective function coefficient of a nonbasic variable x_k is improved by its reduced cost, then the LP will have alternative optimal solutions at least one in which x_k is a basic variable, and at least one in which x_k is not a basic variable.

If the objective function coefficient of a nonbasic variable x_k is improved by more than its reduced cost, then any optimal solution to the LP will have x_k as a basic variable.

Reduced cost of a basic variable is zero!

Shadow Price

We define the shadow price for the i th constraint of an LP to be the amount by which the optimal z value is "improved" (increased in a max problem and decreased in a min problem) if the RHS of the i th constraint is increased by 1.

This definition applies only if the change in the RHS of the constraint leaves the current basis optimal!

A $\geq$ constraint will always have a nonpositive shadow price; a $\leq$ constraint will always have a nonnegative shadow price.

SENSITIVITY ANALYSIS

Conceptualization

$$\max z = 5x_1 + x_2 + 10x_3$$

$$x_1 + x_3 \leq 100$$

$$x_2 \leq 1$$

$$\text{All variables} \geq 0$$

This is a very easy LP model and can be solved manually without utilizing Simplex algorithm. $x_2 = 1$ (This variable does not exist in the first constraint. In this case, as the problem is a maximization problem, the optimum value of the variable equals the RHS value of the second constraint).

$x_1 = 0, x_3 = 100$ (These two variables do exist only in the first constraint and as the objective function coefficient of x_3 is greater than that of x_1 , the optimum value of x_3 equals the RHS value of the first constraint).

Hence, the optimal solution is as follows:

$$z = 1001, [x_1, x_2, x_3] = [0, 1, 100]$$

SENSITIVITY ANALYSIS

Reduced Cost

As x_2 and x_3 are in the basis, their reduced costs are 0.

In order to have x_1 enter in the basis, we should make its objective function coefficient as great as that of x_3 . In other words, improve the coefficient as 5 ($10-5$). New objective function would be ($\max z = 10x_1 + x_2 + 10x_3$) and there would be at least two optimal solutions for $[x_1, x_2, x_3]$: $[0, 1, 100]$ and $[100, 1, 0]$.

Therefore reduced cost of x_1 equals 5.

If we improve the objective function coefficient of x_1 more than its reduced cost, there would be a unique optimal solution: $[100, 1, 0]$.

Shadow Price

If the RHS of the first constraint is increased by 1, new optimal solution of x_3 would be 101 instead of 100. In this case, new z value would be 1011.

If we use the definition: $1011 - 1001 = 10$ is the shadow price of the first constraint. Similarly the shadow price of the second constraint can be calculated as 1 (please find it).

SENSITIVITY ANALYSIS

Some important equations

If the change in the RHS of the constraint leaves the current basis optimal (within the allowable RHS range), the following equations can be used to calculate new objective function value:

for maximization problems

$$\text{new obj. fn. value} = \text{old obj. fn. value} + (\text{new RHS} - \text{old RHS}) \times \text{shadow price}$$

for minimization problems

$$\text{new obj. fn. value} = \text{old obj. fn. value} - (\text{new RHS} - \text{old RHS}) \times \text{shadow price}$$

For this example, as the allowable increases in RHS ranges are infinity for each constraint, we can increase RHS of them as much as we want. But according to allowable decreases, RHS of the first constraint can be decreased by 100 and that of second constraint by 1.

Lets assume that new RHS value of the first constraint is 60.

As the change is within allowable range, we can use the first equation (max. problem):

$$z_{\text{new}} = 1001 + (60 - 100) 10 = 601.$$

SENSITIVITY ANALYSIS

Utilizing Simplex for Sensitivity

In Dakota furniture example; x_1 , x_2 , and x_3 were representing the number of desks, tables, and chairs produced.

The LP formulated for profit maximization:

$$\begin{array}{rcll}
 \max z = & 60 x_1 & 30 x_2 & 20x_3 \\
 & 8 x_1 & + 6 x_2 & + x_3 + s_1 & = 48 & \text{Lumber} \\
 & 4 x_1 & + 2 x_2 & +1.5 x_3 & + s_2 & = 20 & \text{Finishing} \\
 & 2 x_1 & +1.5 x_2 & + .5 x_3 & + s_3 & = 8 & \text{Carpentry} \\
 & & x_2 & & + s_4 & = 5 & \text{Demand}
 \end{array}$$

The optimal solution was:

$$\begin{array}{rcll}
 z & +5 x_2 & +10 s_2 & +10 s_3 & = 280 \\
 & -2 x_2 & +s_1 & +2 s_2 & -8 s_3 & = 24 \\
 & -2 x_2 & + x_3 & +2 s_2 & -4 s_3 & = 8 \\
 + x_1 & + 1.25 x_2 & & -.5 s_2 & +1.5 s_3 & = 2 \\
 & x_2 & & & + s_4 & = 5
 \end{array}$$

SENSITIVITY ANALYSIS

Analysis 1

Suppose available finishing time changes from 20 $\rightarrow$ 20+ δ , then we have the system:

$$\begin{array}{rclcl}
 z' = & 60 x_1' & + 30 x_2' & + 20 x_3' & \\
 & 8 x_1' & + 6 x_2' & + x_3' & + s_1' = 48 \\
 & 4 x_1' & + 2 x_2' & + 1.5 x_3' & + s_2' = 20+\delta \\
 & 2 x_1' & + 1.5 x_2' & + .5 x_3' & + s_3' = 8 \\
 & & + x_2' & & + s_4' = 5
 \end{array}$$

or equivalently:

$$\begin{array}{rclcl}
 z' = & 60 x_1' & + 30 x_2' & + 20 x_3' & \\
 & 8 x_1' & + 6 x_2' & + x_3' & + s_1' = 48 \\
 & 4 x_1' & + 2 x_2' & + 1.5 x_3' & + (s_2' - \delta) = 20 \\
 & 2 x_1' & + 1.5 x_2' & + .5 x_3' & + s_3' = 8 \\
 & & + x_2' & & + s_4' = 5
 \end{array}$$

SENSITIVITY ANALYSIS

That is $z', x_1', x_2', x_3', x_4', s_1', s_2' - \delta, s_3', s_4'$ satisfy the original problem, and hence

(1) Substituting in:

$$\begin{array}{rclclcl}
 z' & +5 x_2' & & +10(s_2' - \delta) & +10 s_3' & = 280 \\
 & -2 x_2' & + s_1' & +2(s_2' - \delta) & -8 s_3' & = 24 \\
 & -2 x_2' & + x_3' & +2(s_2' - \delta) & -4 s_3' & = 8 \\
 + x_1' & +1.25 x_2' & & -.5(s_2' - \delta) & +1.5 s_3' & = 2 \\
 & x_2' & & & & + s_4' = 5
 \end{array}$$

and thus

$$\begin{array}{rclclcl}
 z' & +5 x_2' & & +10 s_2' & +10 s_3' & = 280 + 10\delta \\
 & -2 x_2' & + s_1' & +2 s_2' & -8 s_3' & = 24 + 2\delta \\
 & -2 x_2' & + x_3' & +2 s_2' & -4 s_3' & = 8 + 2\delta \\
 + x_1' & +1.25 x_2' & & -.5 s_2' & +1.5 s_3' & = 2 - .5\delta \\
 & x_2' & & & & + s_4' = 5
 \end{array}$$

For $-4 \leq \delta \leq 4$, the new system maximizes z' . In this range RHS values are non-negative.

As δ increases, revenue increases by 10δ . Therefore, the **shadow price** of finishing labor is \$10 per hr. (This is valid for up to 4 extra hours or 4 fewer hours).

SENSITIVITY ANALYSIS

Analysis 2

What happens if revenue from desks changes to $\$60+\gamma$? For small γ , revenue increases by 2γ (as we are making 2 desks currently). But how large an increase is possible?

The new revenue is:

$$\begin{aligned} z' &= (60+\gamma)x_1 + 30x_2 + 20x_3 = z + \gamma x_1 \\ &= (280 - 5x_2 - 10s_2 - 10s_3) + \gamma(2 - 1.25x_2 + .5s_2 - 1.5s_3) \\ &= 280 + 2\gamma - (5 + 1.25\gamma)x_2 - (10 - .5\gamma)s_2 - (10 + 1.5\gamma)s_3 \end{aligned}$$

So the top line in the final system would be:

$$z' + (5 + 1.25\gamma)x_2 + (10 - .5\gamma)s_2 + (10 + 1.5\gamma)s_3 = 280 + 2\gamma$$

Provided all terms in this row are ≥ 0 , we are still optimal.

For $-4 \leq \gamma \leq 20$, the current production schedule is still optimal.

SENSITIVITY ANALYSIS

Analysis 3

If revenue from a non-basic variable changes, the revenue is

$$\begin{aligned} z' &= 60x_1 + (30 + \gamma)x_2 + 20x_3 = z + \gamma x_2 \\ &= 280 - 5x_2 - 10s_2 - 10s_3 + \gamma x_2 \\ &= 280 - (5 - \gamma)x_2 - 10s_2 - 10s_3 \end{aligned}$$

The current solution is optimal for $\gamma \leq 5$. But when $\gamma > 5$ or the revenue per table is increased past \$35, it becomes better to produce tables. We say the **reduced cost** of tables is \$5.00.